

ABHILASH REDDY SANNAREDDY

Problem-Driven Builder — Product Innovation — AI Systems — Entrepreneurship

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01. Professional Profile

Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning with a strong interest in Product Management, Product Strategy, AI Systems, and Entrepreneurship.

My work focuses on identifying real-world problems, understanding root causes, translating ambiguous challenges into structured product opportunities, and designing technology-enabled solutions. I am particularly interested in the intersection of **problem discovery, AI, product innovation, business processes, and venture building**.

Hands-on experience spans Agentic AI, workflow automation, full-stack development, AI/ML systems, product discovery, business analysis, and startup operations. Experienced in working with founders, students, industry professionals, and multidisciplinary teams to identify problems, define requirements, design solutions, and execute innovation initiatives.

Founder of **Yantriksha X Hub**, a multidisciplinary student innovation ecosystem connecting engineering, management, and law students to problem discovery, product development, entrepreneurship, internships, industry mentorship, and startup opportunities.

02. Career Focus

Product Management	Product discovery, user research, problem framing, requirements gathering, product strategy, prioritization, roadmapping, stakeholder management
AI Product Strategy	Generative AI, Agentic AI, LLM applications, AI workflow automation, enterprise AI, AI-enabled decision systems
Entrepreneurship	Opportunity identification, startup ideation, business-model exploration, venture building, innovation ecosystems
Problem Discovery	Root-cause analysis, unmet-need identification, user-signal analysis, opportunity discovery, market-gap analysis
Innovation	Hackathons, product competitions, interdisciplinary collaboration, student entrepreneurship, industry engagement

03. Education

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science & Technology

B.Tech – Computer Science and Engineering (Artificial Intelligence & Machine Learning) May 2027

CGPA: 9.19/10

Till Semester VI

Academic Areas: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Database Systems, Software Engineering, Full-Stack Development, Cloud Computing, Internet of Things

Academic Focus: AI/ML systems, Agentic AI, software systems, product development, data-driven decision making, and technology-enabled innovation.

04. Professional Experience

Agentic AI Developer Intern

Modern Agriculture Technology Innovation Center (MATIC)

Jun 2025 – Oct 2025

MADeIT Incubated Startup at IITDM Kancheepuram

Chennai, Tamil Nadu, India

Full-Time — Hybrid

Role Overview

Worked with a startup team to identify operational inefficiencies, understand business-process pain points, translate operational problems into product requirements, and design AI-powered automation workflows.

Problem Discovery & Product Work

- Identified fragmented, manual, and inefficient workflows across startup operations that created repetitive work, process delays, information gaps, and unnecessary operational overhead.
- Worked directly with founders and stakeholders to understand operational pain points, investigate root causes, identify high-impact automation opportunities, and translate business problems into structured product requirements.
- Evaluated repetitive operational activities and identified opportunities where AI and workflow automation could reduce manual coordination and improve process efficiency.

AI Systems & Implementation

- Designed AI-driven workflows to automate repetitive operational activities, improve information flow, and reduce dependency on manual coordination.
- Developed and enhanced **Assist Pro**, an AI-powered automation platform that reduced manual effort by **70%+** across startup operations.
- Designed and implemented agentic AI and workflow automation solutions using large language models, workflow orchestration, APIs, and full-stack technologies.
- Converted fragmented manual processes into structured and scalable AI-enabled workflows.
- Participated across problem identification, workflow design, product requirements, implementation, testing, iteration, and operational improvement.

Technology Exposure

Agentic AI — LLMs — Workflow Automation — LangGraph — LangChain — n8n — APIs — Full-Stack Development — AI Workflow Design

05. Leadership & Entrepreneurship

Founder, Chairman & President – Yantriksha X Hub

Student Innovation & Entrepreneurship Ecosystem
Chennai, Tamil Nadu, India

Feb 2025 – Present

Mission

Founded a multidisciplinary student innovation ecosystem focused on helping students identify real-world problems, validate ideas, develop products, explore entrepreneurship, and connect with industry.

Problem Identified

- Identified gaps in interdisciplinary collaboration and structured problem discovery within the student ecosystem, where students often possess technical capabilities but lack structured pathways for validating problems, developing products, understanding business viability, and accessing industry mentorship.
- Designed Yantriksha X Hub as a platform connecting engineering, management, and law students to create multidisciplinary teams around real-world problems.

Innovation Framework

- Developed the **-1 → 0 → 1 framework**, helping students progress from **confusion and ambiguity around a problem (-1)** to a **validated idea (0)** and ultimately toward a **working product (1)**.
- Integrated technical feasibility, business viability, and legal considerations into early-stage product development.
- Used workshops, hackathons, mentorship, project development, and industry interactions as mechanisms for experiential learning.

Scale & Impact

- Built a multidisciplinary ecosystem of **700+ active students** and **300+ alumni**.

- Facilitated **100+ internships** through connections with companies, startups, professionals, and industry networks.
- Supported **100+ patentable projects** and enabled multiple student-led startup initiatives.
- Built a network of **20+ industry experts**.
- Developed collaborations with **IICs, TBIs, startups, and industry partners**.
- Organized innovation programs, hackathons, workshops, mentorship activities, and interdisciplinary initiatives.

Show Director – Just Between Us (JBU)

Student-Led Town Hall Platform
Chennai, Tamil Nadu, India

Feb 2026 – Present

Purpose

Co-created a student-led town hall platform designed to create meaningful conversations between students and accomplished leaders, entrepreneurs, innovators, and public personalities.

- Identified a gap between students and experienced professionals in accessing authentic conversations around leadership, innovation, entrepreneurship, career development, and personal growth.
- Led speaker outreach, content curation, partnership coordination, event planning, audience engagement, and event operations.
- Delivered **4 episodes** featuring **4 accomplished speakers** and engaging **hundreds of students**.

06. Product & Technical Projects

Cloud-Based Restaurant Billing & Order Management Platform

Problem

Local restaurants often rely on conventional billing and POS infrastructure requiring dedicated computers, hardware, maintenance, and recurring technology costs.

Product Concept

Designed a low-cost cloud-based billing and order management platform integrating billing, order processing, delivery workflows, and thermal printing into a unified ecosystem.

Technical Solution

- Built an ESP-based thermal printing system using MQTT, enabling thermal printers to receive and process orders without requiring a dedicated computer at each printer.
- Designed a cloud-connected architecture allowing order information to move from the software platform to the printer through a lightweight communication layer.
- Integrated restaurant billing, order processing, delivery workflows, and thermal printing.

Product / Business Outcome

- Reduced hardware and maintenance requirements for restaurants.
- Simplified restaurant order processing and reduced missed-order situations.
- Enabled pilot operations to scale from approximately **25 orders/day** to **100–150+ orders/day**.

Technology: Cloud Architecture — MQTT — ESP — Thermal Printing — Full-Stack Development

Precision Bid Management & Tender Analysis System

Context

Developed as a solution concept during the Pragyan Hackathon in collaboration with Aurigo Software Technologies.

Problem

Contractors often rely on manual evaluation of technical requirements, financial information, competitor pricing, historical bids, material costs, and bidding trends when preparing tender submissions.

Product Approach

- Designed an AI/ML-powered decision-support platform for tender and bid analysis.
- Analyzed historical bids, contracts, bidding trends, competitor pricing, material costs, and technical and commercial parameters.
- Used historical bidding information to identify patterns in winning bid prices, competitor pricing trends, material costs, and bidding behavior.
- Designed the decision-support process to help contractors balance competitiveness, probability of winning, and target profitability.

Technology / Methods: Machine Learning — Historical Data Analysis — Decision Support — Competitive Analysis

AI-Powered Smart Agriculture & Farm-to-Market Platform

Recognition: Winner – Prototyping Contest

Problem

Farmers often manage seed procurement, crop planning, irrigation, farm monitoring, and produce selling through fragmented processes.

Product Concept

Designed an integrated farm-to-market platform connecting:

Seed Procurement → Crop Planning → Farm Monitoring → AI Irrigation → Produce Selling

Technical Approach

- Integrated IoT sensors to monitor agricultural parameters including NPK levels, soil moisture, temperature, humidity, and pH.
- Applied machine learning to agricultural data for crop recommendations and irrigation decisions.
- Designed an integrated product ecosystem rather than treating procurement, farm monitoring, irrigation, and selling as separate workflows.
- Achieved **98.4% prediction accuracy** in the developed prediction system.

07. Product Strategy & Case Studies

The following are product strategy, research, and business-design case studies. They represent analysis, product design, strategic proposals, and conceptual frameworks rather than deployed products.

AI Salesperson for E-Commerce

Product Strategy Case Study

[Detailed Case Study]

Problem

E-commerce has made product discovery highly efficient, but customers still perform much of the product decision-making themselves. Customers must translate their needs into search queries, determine which specifications matter, apply filters, compare alternatives, interpret technical information, and evaluate trade-offs.

The underlying gap identified was not simply product discovery. It was the lack of **deep customer-intent understanding**.

Product Thesis

The product opportunity is to recreate the intelligence of a good salesperson in a digital environment.

Proposed Product

- Designed an AI salesperson that proactively asks decision-relevant questions rather than waiting for customers to specify every requirement.
- Converts natural-language needs into structured product requirements.
- Uses customer preferences and rejection feedback to dynamically refine recommendations.
- Narrows large product catalogs into personalized shortlists rather than presenting customers with excessive options.
- Explores product intelligence and cross-marketplace comparison across authorized commerce sources.

Strategic Positioning

Envisioned the product as a **Customer Decision Engine for Commerce** rather than another chatbot, search engine, marketplace, or price-comparison tool.

The proposed transformation is:

Search → Filters → Lists → Research → Decision

to:

Conversation → Understanding → Recommendation → Confidence → Purchase

Delivery ETA Intelligence Platform

Product Strategy Case Study

[Detailed Case Study]

Problem

Customers often receive broad delivery windows such as “Out for Delivery” or “Delivery by End of Day” even when logistics providers already possess route, rider, GPS, traffic, package-sequence, and delivery-completion information.

This creates customer uncertainty and can contribute to WISMO support requests, failed delivery attempts, re-delivery costs, and lower customer satisfaction.

Root Cause

The identified gap was not the absence of logistics intelligence.

The problem was that operational visibility remained primarily internal instead of being translated into useful customer-facing delivery predictions.

Proposed Product

Designed an ETA intelligence layer:

Internal Logistics Data → Prediction Engine → Customer Delivery Window

Potential inputs include:

- Route assignment
- Rider allocation
- Package sequence
- GPS location
- Traffic conditions
- Historical route data
- Delivery completion events
- Failed delivery patterns

Product Strategy

Proposed a phased strategy:

1. Smart delivery slots
2. Dynamic ETA updates
3. Proactive customer notifications
4. Delivery intelligence dashboard for logistics partners

Success metrics include ETA accuracy, WISMO reduction, failed-delivery reduction, customer satisfaction, notification engagement, and re-delivery savings.

OpportunityOS – Agentic AI Product Opportunity Discovery Framework

Product Discovery Research / Conceptual Framework

[Detailed Case Study]

Research Question

How can heterogeneous user signals be continuously transformed into validated product opportunities?

Problem

Organizations receive massive volumes of signals through app reviews, social media, developer forums, support tickets, search trends, and community discussions. These signals contain information about frustrations, unmet needs, failures, and emerging opportunities but remain fragmented, noisy, and difficult to synthesize into product opportunities.

Core Concept

Designed **OpportunityOS**, a conceptual agentic AI framework that moves from:

Raw Signals → Problems → Opportunities

rather than beginning with an already-defined problem.

Conceptual Architecture

The proposed architecture contains five layers:

1. Signal Ingestion
2. Signal Processing & Normalisation
3. Agentic Problem Extraction
4. Opportunity Synthesis & Scoring
5. Opportunity Intelligence Layer

Agentic Workflow

Designed specialized agents for:

- Signal extraction
- Validation and evidence grounding
- Problem clustering
- Opportunity synthesis
- Opportunity scoring

Opportunity Scoring

Proposed an auditable opportunity-scoring framework across:

Severity — Prevalence — Growth Momentum — Underservedness — Feasibility — Novelty

Research Perspective

The framework explores the transition from **problem-solving intelligence** to **problem-discovery intelligence**, while retaining human decision-makers as the final authority in opportunity selection.

Intics – Strategic Product Analysis

Enterprise AI Product Strategy

[Detailed Case Study]

Strategic Question

How can enterprise AI move beyond generating recommendations toward measurable business outcomes, workflow execution, trusted decisions, and broader operating intelligence?

Primary Strategic Gap: Outcome Intelligence

Identified a strategic gap where enterprise AI recommendations can become the endpoint of a workflow, leaving organizations without a systematic mechanism to determine whether those decisions actually produced successful business outcomes.

Proposed Outcome Intelligence Layer

Proposed connecting:

Document → Decision → Outcome → Learning → Improved Future Decisions

Potential outcome signals include:

- Loan performance
- Appeal outcomes
- Bid win rates
- Project performance
- Compliance outcomes
- Reviewer overrides
- Operational outcomes

Broader Strategic Exploration

Explored the evolution of enterprise intelligence across:

Document Intelligence → Decision Intelligence → System of Action → Enterprise Operating Intelligence

The analysis considered outcome measurement, workflow execution, explainability, trust, data sovereignty, and predictive operational intelligence.

India Battery Lifecycle & Recovery Platform

Startup Strategy / Circular Economy Case Study

[Detailed Case Study]

Problem

Identified fragmentation in the end-of-life EV battery lifecycle. Old, damaged, and warranty-replaced batteries can originate from OEM service centres, dealers, fleets, workshops, accident or insurance channels, and other scattered locations.

The central product question is:

Who collects the battery, assesses its remaining value, determines its appropriate destination, and tracks the lifecycle?

Market Gap

The opportunity is not positioned as simply building another recycling company. Existing players already operate across parts of collection, recycling, EPR, and reverse logistics.

The identified opportunity is the **missing middle** between fragmented battery sources and downstream processing.

Proposed Product Model

Find → Collect → Identify → Test/Assess → Grade → Aggregate → Track → Route

Routing Logic

- Potentially reusable / refurbishable → Authorised refurbisher
- Suitable for validated second-life application → Appropriate second-life partner
- Genuinely end-of-life → Authorised recycler
- Damaged / unsafe → Appropriate specialised handling

Business Strategy

Proposed an initially neutral and asset-light model that partners with existing recyclers and refurbishers rather than building recycling infrastructure immediately.

The platform would create a collection, assessment, grading, aggregation, traceability, and routing layer for OEMs, fleets, dealers, service centres, recyclers, and refurbishers.

Long-term opportunities include battery intelligence, digital traceability, second-life applications, recycling infrastructure, and broader battery circular-economy infrastructure.

08. Research & Innovation

Opportunity Discovery, Agentic AI & Product Innovation

Research and conceptual work focused on the emerging problem of identifying **what should be built** rather than only optimizing how known solutions are built.

Key areas explored:

- Problem discovery intelligence
- Agentic AI and multi-agent orchestration
- User-signal mining
- Product opportunity identification
- Requirements mining
- Weak-signal detection
- Retrieval-grounded reasoning
- Opportunity evaluation
- Human-in-the-loop product decision making

Enterprise AI Strategy

Strategic analysis explored how enterprise AI platforms can evolve across:

- Document Intelligence
- Decision Intelligence
- Outcome Intelligence
- System of Action
- Trusted Decision Intelligence
- Enterprise Operating Intelligence

Focus areas included business outcomes, decision quality, explainability, trust, workflow execution, data sovereignty, and predictive operational intelligence.

09. Achievements & Competitions

VISAI 2026 – Organizer

16th International Project Competition & Exhibition

- Coordinated a multi-stakeholder innovation event involving **720+ students, 240+ teams, 44 institutions, and 14 industry partners**.
- Supported participant coordination, innovation activities, industry engagement, and event execution.

Smart India Hackathon 2026 – Internal Hackathon Organizer

- Streamlined the internal participation process for Smart India Hackathon 2026.
- Digitally engaged and motivated students through Yantriksha X Hub.
- Helped scale participation from **70 teams to 200+ teams**.
- Facilitated connections between participating teams and mentors to support idea development and execution.

AI-Powered Smart Agriculture & Farm-to-Market Platform

- **Winner – Prototyping Contest**
- **Runner-Up – Project Idea Contest**

Cybersecurity Bootcamp

- **Winner – Cybersecurity Bootcamp, IIITDM Kancheepuram.**

Innovation Marathon Evaluator

- **KRM Public School**
- Evaluated **300+ student innovation and project submissions**.

L&T Techgium Hackathon

- Organizer – Preliminary Rounds of the L&T Techgium Hackathon.

SDIP 4.0

- Volunteer – SDIP 4.0, organized by **EDII-Tamil Nadu, Government of Tamil Nadu**.

10. Community & Industry Engagement

Student Innovation Ecosystem

- Built connections between students, startup founders, industry professionals, mentors, innovation cells, technology business incubators, and entrepreneurial communities.
- Created opportunities for students to interact with professionals across technology, product development, entrepreneurship, innovation, and business.
- Organized workshops, hackathons, project competitions, mentorship sessions, innovation programs, and entrepreneurship activities.

Industry Interaction

- Engaged with industry experts through Yantriksha X Hub programs and initiatives.
- Facilitated mentor and professional connections for students participating in hackathons and innovation programs.
- Supported student teams in connecting technical ideas with business, product, and execution perspectives.

11. Technical & Product Skills

Product Management

Product Discovery, User Research, Problem Framing, Requirements Gathering, Product Strategy, Market Research, Competitive Analysis, Product Roadmapping, Feature Prioritization, Agile, Stakeholder Management

Business & Analytics

Business Analysis, Process Optimization, Go-to-Market Strategy, SQL, Microsoft Excel, Power BI, KPI Tracking, Data-Driven Decision Making

AI & Agentic Systems

Machine Learning, Generative AI, Large Language Models, Agentic AI, LangGraph, LangChain, LangSmith, AI Workflow Automation, n8n, Prompt Engineering, Internet of Things

Software Development

Python, Java, Spring Framework, JavaScript, React.js, Node.js, HTML, CSS

Cloud & Infrastructure

Amazon Web Services (AWS), Google Cloud Platform (GCP), Firebase

Product & Collaboration Tools

Jira, Notion, Figma, Git, GitHub

12. Languages

English	Professional Proficiency
Telugu	Native Proficiency
Hindi	Professional Proficiency
Tamil	Working Proficiency
Kannada	Basic Proficiency
German	Basic Proficiency

13. Areas of Interest

Product Management	AI Product Strategy
Agentic AI	Product Discovery
Entrepreneurship	Venture Building
Enterprise AI	Business Process Automation
Innovation Ecosystems	Technology Entrepreneurship
AI/ML Applications	Product Strategy

14. Selected Portfolio

- **AI Salesperson for E-Commerce:** Product strategy and customer decision-engine design.
- **Delivery ETA Intelligence Platform:** Last-mile logistics product strategy and customer-facing ETA intelligence.
- **OpportunityOS:** Agentic AI framework for automated product opportunity discovery.
- **Intics Strategic Product Analysis:** Enterprise AI outcome intelligence and strategic evolution.
- **India Battery Lifecycle & Recovery Platform:** Battery lifecycle infrastructure and circular-economy startup strategy.